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DESIGN AN IMPLEMENTATION OF IRS FOR SMALL MIMO WIRELESS NETWORK

A PROJECT REPORT

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ABSTRACT

The **exponential growth of wireless data demand** and the **limitations of conventional communication infrastructure** have necessitated the development of **innovative technologies** to improve **signal quality** and **energy efficiency**. **Intelligent Reflecting Surfaces (IRS)** have emerged as a **promising solution** in **next-generation wireless networks**.

This project explores the **design and implementation** of an **IRS-assisted wireless communication system**, where a **programmable meta surface** is used to **dynamically control** the reflection of incident signals toward the receiver, thereby **enhancing the overall communication performance**.

The **IRS is modelled using discrete phase shifts** and is integrated into a **simulation environment** to analyse its impact on system parameters such as **signal-to-noise ratio (SNR)**, **bit error rate (BER)**, and **achievable data rate**. **Optimization algorithms** are employed to configure the IRS elements in real-time, **maximizing signal strength** at the receiver while **minimizing interference**.

The project demonstrates that **IRS can significantly improve wireless coverage, reliability, and spectral efficiency**, especially in **non-line-of-sight (NLoS) scenarios**.

This work contributes toward the **practical realization of IRS in 6G networks**, highlighting its potential in **future smart radio environments**.

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ABBREVIATIONS

S. NO.	ABBREVIATION	EXPANSION
1	IRS	Intelligent Reflecting Surface
2	MIMO	Multiple Input Multiple Output
3	IoT	Internet Of Things
4	NLOS	Non-Line-Of-Sight
5	SNR	Signal-to-Noise Ratio
6	FPGA	Field-Programmable Gate Array
7	RF	Radio Frequency
8	WSN	Wireless Sensor Network
9	CSI	Channel State Information
10	AI	Artificial Intelligence
11	ML	Machine Learning
12	WPCN	Wireless Powered Communication Network
13	ISWPT	Integrated Sensing and Wireless Power Transfer
14	SDP	Semi-Definite Programming
15	LC	Low Complexity
16	MM	Majorization-Minimization
17	HAP	High-Altitude Platform
18	WD	Wireless Devices
19	AR/VR	Augmented Reality / Virtual Reality
20	FEA	Finite Element Analysis
21	GUI	Graphical User Interface
22	EMC	Electro-Magnetic Compatibility
23	MEMS	Micro-Electro-Mechanical Systems
24	SISO	Single Input Single Output
25	OFDM	Orthogonal Frequency Division Multiplexing

26	URLLC	Ultra-Reliable Low-Latency Communication
27	Wi-Fi	Wireless Fidelity
28	AM	Amplitude Modulation
29	FM	Frequency Modulation
30	PSK	Phase Shift Keying
31	FSK	Frequency Shift Keying
32	APES	Amplitude and Phase Estimation